

Neuro-Symbolic Fleet Dispatching: Waynex Engine

Author: Sharath Kumar Reddy Kapu | Email: skrkapu@gmail.com
GitHub: <https://github.com/mr-sharath/DRIFT/> | Web: <https://waynex.up.railway.app/>

ABSTRACT:

Capacitated Vehicle Routing Problems (CVRP) in modern urban logistics are traditionally solved using MILP techniques or metaheuristic local search solvers like Google OR-Tools. In this paper, we present Waynex, a neuro-symbolic fleet optimization engine combining Spatial GNN encoder with Advantage Actor-Critic (A2C) DRL policy and 2-Opt refinement. Across synthetic graphs (N=10 to N=200) and real urban topologies (Bengaluru, SF, Tokyo), Waynex OPT2 achieves 1.39ms to 116.24ms inference latencies (8.6x to 721x speedup).

1. SYSTEM ARCHITECTURE:

- 1. Spatial GNN Encoder: Maps 6D node features into 64D latent node embeddings.
- 2. A2C Policy Network with Action Masking: Dynamically enforces capacity constraints.
- 3. Symbolic 2-Opt Local Search: Untangles path intersections in sub-millisecond time.

2. STATISTICAL BENCHMARKS (Mean +/- Std Over 5 Random Seeds):

Nodes (N)	OR-Tools Time (ms)	Waynex v2 Time (ms)	OR-Tools Dist (km)	Waynex v2 Dist
10	1003.02 +/- 4.34	1.39 +/- 0.08	74.80 +/- 9.45	68.34 +/- 6.54
20	1001.20 +/- 0.16	3.42 +/- 0.22	99.48 +/- 5.40	102.31 +/- 9.58
50	1001.28 +/- 0.37	10.99 +/- 0.55	199.80 +/- 10.70	257.30 +/- 3.62
100	1002.07 +/- 0.05	31.20 +/- 1.06	291.92 +/- 19.11	534.77 +/- 18.04
200	1004.49 +/- 0.28	116.24 +/- 9.05	445.77 +/- 18.38	1037.28 +/- 48.62

Real Urban Topologies:

- Bengaluru: Waynex OPT2 1.47ms (210.79 km) vs OR-Tools 1005.28ms (213.61 km) -> 683x Speedup
- San Francisco: Waynex OPT2 1.38ms (61.56 km) vs OR-Tools 1000.89ms (61.78 km) -> 725x Speedup
- Tokyo Metro: Waynex OPT2 1.05ms (68.30 km) vs OR-Tools 1000.95ms (82.23 km) -> 953x Speedup

3. REPRODUCIBILITY & HARDWARE SPECS:

Executed on Apple M-series CPU (1 vCPU thread), Python 3.11, PyTorch 2.5.1, OR-Tools 9.8.
Source code & weights: <https://github.com/mr-sharath/DRIFT/>